

DEPARTMENT OF SOE
MASTER OF COMPUTER APPLICATIONS (MCA)

Final Year Project — Semester III (2026 - 2027)

Literature Review & Research Understanding Report
(Part II — 10 Advanced Research Papers)

ZeroTrustCloud / TrustMesh

Investigation of Cloud-Native Identities, Service Mesh Hardening, SGX Enclaves, Network Misconfigurations,
and Dynamic Risk-Based Access Control

Project Domain: Cybersecurity & Cloud-Native Security

Proposed Platform: TrustMesh (Identity-Centric Zero-Trust Control Plane)

Research Phase: Phase II Literature Analysis & Technical Scoping

SUBMITTED BY

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1. Project Research Context & Phase II Review Scope

In Phase I of our literature survey, our team analyzed 10 foundational papers examining the macro-principles of Zero-Trust Architecture (ZTA), OIDC-based workload identity federation, Kubernetes Service Account binding, baseline mTLS overhead in service meshes, and CI/CD software supply chain integrity.

While Phase I established the broad theoretical necessity of 'never trust, always verify', it revealed several acute architectural questions that require deeper scientific grounding before constructing our proposed platform, TrustMesh:

- **Vulnerability-Aware Access Control:** Standard service mesh proxies enforce binary authentication based on cryptographic certificates (x509 SVIDs). If a microservice has known high-severity CVEs in its third-party dependencies, how can the mesh dynamically restrict or quarantine its privileges without manual intervention?
- **The Threat of Malicious Control Planes & Untrusted Hosts:** Existing Zero-Trust models assume the control plane (e.g., istiod, Kubernetes API server) and host operating systems are entirely trustworthy. What happens if a cluster administrator is compromised or host root is breached? Can cryptographic signing or hardware-assisted enclaves (Intel SGX) provide verifiable policy execution?
- **Empirical Reality of Kubernetes Network Misconfigurations:** How severe are real-world lateral movement vulnerabilities in production Kubernetes clusters, and how can automated least-privilege network policies be inferred and enforced?
- **Multi-Cloud Performance Reality:** What are the exact latency distributions, standard deviations (jitter), and resource footprints of running service mesh proxies across multi-cloud environments like AWS EKS and Azure AKS?
- **Elimination of Static Bootstrapping Secrets:** Can inter-application and cross-cloud communication bootstrap securely using cloud-native managed identities without hardcoding any long-lived API tokens or TLS certs?

To address these critical questions, this second literature review examines 10 advanced, peer-reviewed research papers published between 2024 and 2026. This report provides a comprehensive breakdown of each paper: theoretical problem, methodology, empirical findings, direct relevance to TrustMesh, concrete implementation plans, and identified technical gaps.

Paper 1 — A Simpler Approach to Inter-Application Authentication Utilising Cloud-Native Identities

Paper Details

- **Title:** A Simpler Approach to Inter-Application Authentication Utilising Cloud-Native Identities
- **Authors:** Asher and Muttukrishnan Rajarajan (City, University of London, UK)
- **Publication / Venue:** Journal of Cloud Computing (Springer Open), Vol. 15, Article 83, February 2026. DOI: 10.1186/s13677-026-00903-8
- **Domain:** Cloud-Native Identity, Secretless Inter-Service Authentication, Ephemeral JWTs

Problem Addressed

Modern cloud-native microservices frequently authenticate to peer services and external APIs using static credentials such as long-lived API tokens, mutual TLS client certificates, and hardcoded database passwords. These static credentials represent persistent pre-granted trust—a direct violation of Zero Trust principles. They frequently leak into version control repositories or package registries (e.g., the 2022 Toyota leak where access keys were exposed on GitHub for five years, and the 2022 InfoSys administrative AWS key leak on PyPI). In large-scale multi-cloud systems, rotating static credentials without application downtime is nearly impossible.

Approach / Methodology & Architecture

The authors propose a cloud-agnostic inter-application authentication framework that completely eliminates static secrets by leveraging cloud-native bootstrap identities (such as AWS EC2/EKS IAM roles, Azure Managed Identities, and GCP Service Accounts). The caller requests a cryptographically signed identity token from its native cloud metadata service, which is presented to an intermediary organizational Identity Provider (IdP) or directly to the target service. The target validates the token against the cloud provider's public OIDC discovery keys and issues a short-lived JSON Web Token (JWT) scoped specifically to the destination API.

What We Understood (Key Findings & Insights)

- **Secretless Inception:** Workloads do not require any bootstrap secret injected at startup; their runtime compute identity serves as the cryptographic root.
- **Audience Scoping Mitigates Confused Deputy:** By enforcing strict 'aud' (audience) claim binding in ephemeral JWTs, a token issued for Service A cannot be replayed by an attacker against Service B.
- **Performance Overhead is Negligible with Token Caching:** The initial authentication handshake takes 18-35 ms, but subsequent requests using cached ephemeral tokens operate with sub-millisecond overhead (0.12 ms).
- **Comparison with AWS Outbound Federation:** While AWS Outbound Federation is tightly coupled to AWS STS, the authors' intermediary IdP approach enables seamless interoperability across heterogeneous clouds.

Relevance to Our Project (TrustMesh)

This paper forms the architectural foundation for TrustMesh's secretless service-to-service communication. It solves the cold-start problem of how our microservices on AWS EKS and Azure AKS authenticate to each other and to the TrustMesh control plane without embedding static secrets into Helm charts or Kubernetes Secrets.

What We Plan to Apply

- **Eliminate static secrets:** Ban all static API keys and passwords in TrustMesh microservice deployments.
- **Cloud metadata binding:** Use AWS IRSA and Azure Workload Identity to obtain ephemeral bootstrap tokens.
- **Enforce short-lived JWTs:** Set maximum token lifetimes to 15 minutes with automated in-memory rotation.

Limitations / Identified Research Gaps

The paper focuses strictly on authentication (AuthN); it does not provide fine-grained Layer-7 authorization policies (AuthZ) or integrate Policy-as-Code. Furthermore, it assumes the calling workload itself is uncompromised and does not verify runtime container integrity or software supply chain provenance.

Paper 2 — Decentralised Identity Management Solution for Zero-Trust Multi-Domain Computing Continuum Frameworks

Paper Details

- **Title:** Decentralised Identity Management Solution for Zero-Trust Multi-Domain Computing Continuum Frameworks
- **Authors:** José Manuel Bernabé Murcia, Eduardo Cánovas, Jesús García-Rodríguez, Alejandro M. Zarca, Antonio Skarmeta
- **Affiliation:** University of Murcia & University Center of Defense at the Spanish Air Force Academy, Spain
- **Publication / Venue:** Future Generation Computer Systems (Elsevier), Vol. 160, pp. 1-16, 2024/2025. DOI: 10.1016/j.future.2024.08.012
- **Domain:** Self-Sovereign Identity (SSI), W3C Decentralized Identifiers (DIDs), Verifiable Credentials (VCs)

Problem Addressed

In the Computing Continuum (spanning edge devices, fog nodes, and multi-tenant clouds), traditional centralized Identity Providers (IdPs) like Active Directory or Okta become critical bottlenecks and single points of failure. Furthermore, cross-organizational and cross-cloud collaboration suffers from identity provider lock-in, lack of interoperability, and privacy concerns when user/workload attributes must be shared across untrusted administrative boundaries.

Approach / Methodology & Architecture

The authors design a Decentralised Identity Management (DIM) framework based on Self-Sovereign Identity (SSI) and W3C standards: Decentralized Identifiers (DIDs) and Verifiable Credentials (VCs). Using a Distributed Ledger Technology (DLT) anchor (Hyperledger Indy) and Aries agents, entities hold cryptographically verifiable credentials containing their attributes and capabilities. Access decisions are evaluated dynamically using a capability-based authorization engine without querying a central authentication server at runtime.

What We Understood (Key Findings & Insights)

- **Zero-Knowledge Attribute Verification:** Entities can prove entitlement (e.g., 'authorized for cluster management') using zero-knowledge proofs without revealing underlying sensitive identifiers.
- **Elimination of Central IdP Bottlenecks:** Verification relies on public cryptographic keys anchored on the ledger, enabling offline and cross-domain verification in 4.2 ms.
- **Credential Issuance & Presentation Latency:** Issuing a verifiable credential takes ~180 ms, while presentation verification takes ~12 ms, making it highly viable for workload attestation.
- **Revocation Management:** Cryptographic accumulator revocation lists allow real-time credential revocation without relying on slow OCSP or CRL polling.

Relevance to Our Project (TrustMesh)

TrustMesh operates across multi-cloud clusters (AWS EKS and Azure AKS). Relying exclusively on AWS IAM or Azure AD creates cross-cloud dependency deadlocks. The verifiable credential concepts from this paper provide the blueprint for federating workload attestation across disparate cloud domains.

What We Plan to Apply

- **Decoupled Identity Attestation:** Model TrustMesh workload identities as cryptographically verifiable claims rather than proprietary cloud IAM objects.

- **Cryptographic Credential Revocation:** Implement real-time revocation tokens for microservices flagged by runtime security checks.

Limitations / Identified Research Gaps

The underlying DLT ledger introduces operational complexity that is excessive for high-throughput, short-lived microservice calls (thousands of RPS). In containerized microservices, credential lifecycles are measured in seconds, making blockchain confirmation times an impedance mismatch.

Paper 3 — Enhancing Microservice Security Through Vulnerability-Driven Trust in the Service Mesh Architecture

Paper Details

- **Title:** Enhancing Microservice Security Through Vulnerability-Driven Trust in the Service Mesh Architecture
- **Authors:** Rami Alboqmi and Rose F. Gamble
- **Affiliation:** Tandy School of Computer Science, University of Tulsa, Oklahoma, USA
- **Publication / Venue:** Sensors (MDPI), Vol. 25, Issue 3, Article 914, February 2025. DOI: 10.3390/s25030914
- **Domain:** Service Mesh Security, Vulnerability-Driven Dynamic Trust, Istio / Envoy, Supply Chain Security

Problem Addressed

Service meshes (such as Istio and Linkerd) enforce mutual TLS (mTLS) and authenticate microservices via x509 cryptographic certificates. However, they suffer from a fundamental blind spot: certificate-based trust is completely binary. A microservice with a valid certificate is granted full mesh communication privileges, even if its container image contains critical known vulnerabilities (CVEs) in open-source dependencies or supply-chain packages. An attacker exploiting an unpatched remote code execution (RCE) flaw can use the compromised container's legitimate mTLS certificate to move laterally across the entire cluster.

Approach / Methodology & Architecture

The authors propose the Vulnerability-Driven Trust (VDT) architecture for service meshes. VDT continuously inspects the Software Bill of Materials (SBOM) of running microservices and tracks published CVEs via the National Vulnerability Database (NVD). Using the Common Vulnerability Scoring System (CVSS v3.1), VDT dynamically computes a quantitative Trust Score (ranging from 0.0 to 1.0) for each microservice. When vulnerabilities are detected or threat scores cross predefined thresholds, VDT automatically updates Istio AuthorizationPolicies in real time, degrading permissions or quarantining the vulnerable pod.

What We Understood (Key Findings & Insights)

- **Dynamic Trust Score Formulation:** Trust is calculated as a composite function of exploitability metrics (Attack Vector, Attack Complexity, Privileges Required, User Interaction) and impact metrics (Confidentiality, Integrity, Availability).
- **Automated Policy Adaptation:** The service mesh moves beyond static access control to adaptive quarantine: a pod with a critical CVSS ≥ 9.0 is automatically restricted from communicating with sensitive backend databases.
- **Evaluation on Microservice Benchmark:** Tested on Google's Online Boutique microservice benchmark; dynamic policy generation completed in under 45 ms, and sidecar policy updates executed without dropping active TCP connections.
- **Closing the CI/CD-to-Runtime Gap:** Vulnerability scanning is transformed from a static, one-time build gate into a continuous runtime control plane input.

Relevance to Our Project (TrustMesh)

This paper is a direct, groundbreaking contribution to TrustMesh's Continuous Verification & Supply Chain Engine. It bridges the exact gap identified in our problem statement: connecting SBOM/vulnerability data directly to Istio Layer-7 AuthorizationPolicies.

What We Plan to Apply

- **Continuous Trust Scoring:** Implement a lightweight Trust Score daemon in TrustMesh evaluating container vulnerability profiles.
- **Adaptive Istio Authorization:** Configure OPA and Istio controllers to dynamically revoke service access when container vulnerability thresholds are exceeded.
- **Dynamic Pod Quarantine:** Automatically restrict vulnerable microservices to a segregated network namespace until patched.

Limitations / Identified Research Gaps

The approach depends on external NVD vulnerability feeds and known CVEs. It cannot detect zero-day exploits or runtime behavioral anomalies (e.g., unauthorized process execution or abnormal syscalls) where no CVE yet exists.

Paper 4 — Implementing a Zero Trust Architecture in Hybrid Cloud Environments

Paper Details

- **Title:** Implementing a Zero Trust Architecture in Hybrid Cloud Environments
- **Author:** Phani Sekhar Emmanni (Technical Project Manager, Arizona, USA)
- **Publication / Venue:** International Journal of Computer Trends and Technology (IJCTT), Vol. 72, Issue 5, pp. 33-39, May 2024. DOI: 10.14445/22312803/IJCTT-V72I5P104
- **Domain:** Hybrid Cloud Security, Micro-segmentation, Software-Defined Perimeter (SDP), Zero-Trust IAM

Problem Addressed

Organizations migrating to hybrid cloud infrastructures face severe operational friction when attempting to maintain uniform security posture across on-premises data centers and public cloud platforms. Traditional perimeter defenses fail because data and workloads flow dynamically across heterogeneous network boundaries, leading to inconsistent access control policies, fragmented audit logs, and enlarged attack surfaces.

Approach / Methodology & Architecture

The paper synthesizes enterprise case studies and formulates an implementation framework for hybrid cloud Zero Trust. The architecture emphasizes three core pillars: (1) Identity and Access Management with context-aware Multi-Factor Authentication (MFA) and least privilege; (2) Micro-segmentation utilizing Software-Defined Perimeters (SDP) to create logical isolation down to individual workloads; and (3) Centralized Policy Decision Points (PDP) coupled with distributed Policy Enforcement Points (PEP) and continuous telemetry monitoring.

What We Understood (Key Findings & Insights)

- **Granular Micro-segmentation:** Replacing broad network VLANs with application-level micro-segmentation reduces the blast radius of lateral movement by up to 78% in analyzed enterprise case studies.
- **Separation of PDP and PEP:** Centralizing policy decisions while distributing policy enforcement at edge proxies prevents latency bottlenecks while ensuring uniform policy application.
- **Identity as the New Perimeter:** Network topology must be treated as untrusted transport; identity attributes (user role, device posture, location, time) dictate access.
- **Phased Migration Strategy:** Zero-Trust cannot be deployed overnight; organizations must follow an iterative model: Discovery -> Identity Baselines -> Micro-segmentation -> Automated Policy Enforcement.

Relevance to Our Project (TrustMesh)

Provides architectural design patterns for TrustMesh's hybrid and multi-cloud control plane. Specifically, the separation of PDP (our centralized OPA engine) from PEPs (Envoy sidecars and Kubernetes admission controllers) validates our system blueprint.

What We Plan to Apply

- **PDP/PEP Architecture:** Structure TrustMesh with centralized policy governance and localized sidecar policy enforcement.

- **Phased Verification:** Adopt the paper's phased rollout methodology for testing TrustMesh modules iteratively.

Limitations / Identified Research Gaps

The paper is predominantly conceptual and architectural; it lacks quantitative empirical benchmark experiments and does not address Kubernetes-specific primitives (e.g., CRI, CNI, eBPF) or CI/CD supply-chain signing.

Paper 5 — Inside Job: Defending Kubernetes Clusters Against Network Misconfigurations

Paper Details

- **Title:** Inside Job: Defending Kubernetes Clusters Against Network Misconfigurations
- **Authors:** Jacopo Bufalino, Jose Luis Martin-Navarro, Mario Di Francesco, Tuomas Aura
- **Affiliation:** CNAM (France), Universitat Politècnica de València (Spain), and Aalto University (Finland)
- **Publication / Venue:** ACM Transactions on Privacy and Security / ACM Conference Reference Format, 2025
- **Domain:** Kubernetes Network Security, NetworkPolicy Misconfigurations, Lateral Movement Defense

Problem Addressed

Kubernetes implements a default 'flat network' architecture where all pods can communicate with all other pods across the entire cluster without restriction. To restrict traffic, administrators must write declarative NetworkPolicies. However, because NetworkPolicy syntax is complex and error-prone, developers either omit them entirely or configure overly permissive rules. Consequently, if an attacker compromises a single public-facing pod, they can easily launch lateral attacks ('inside job') against internal databases, monitoring services, and cluster infrastructure.

Approach / Methodology & Architecture

The authors conducted the first large-scale empirical audit of Kubernetes network configurations across 287 open-source application deployments and Helm charts from 6 major organizations (including Bitnami, GitLab, and public sector repositories). They identified 634 critical network misconfigurations. To resolve this, they developed an automated defensive toolchain that performs static analysis on Kubernetes manifests and Helm templates, generates application dependency graphs, and synthesizes least-privilege NetworkPolicies that block unauthorized lateral movement while preserving legitimate application traffic.

What We Understood (Key Findings & Insights)

- **Alarming Prevalence of Misconfigurations:** Over 87% of surveyed production applications lacked adequate network isolation; internal services (e.g., Redis, PostgreSQL) were left completely exposed to all cluster namespaces.
- **Cross-Namespace Exposure:** Default ClusterIP services allow arbitrary cross-namespace traffic unless explicitly restricted by ingress NetworkPolicies.
- **Automated Policy Synthesis:** Static manifest inspection can reliably infer required inter-pod communication ports, enabling automated generation of deny-all default NetworkPolicies with explicit allow-lists.
- **Responsible Disclosure & Fixes:** The authors' synthesized patches were merged into over 30 major open-source application charts, proving the practical necessity of automated policy tooling.

Relevance to Our Project (TrustMesh)

This paper reveals an essential truth for TrustMesh: service mesh mTLS alone is insufficient if the underlying Kubernetes Layer-3/4 CNI allows unrestricted network connections! TrustMesh must enforce automated, least-privilege Kubernetes NetworkPolicies alongside Layer-7 Istio authorization.

What We Plan to Apply

- **Default Deny-All NetworkPolicies:** Enforce a cluster-wide baseline policy rejecting all inter-namespace traffic by default.
- **Automated Policy-as-Code Ingress Rules:** Use OPA Gatekeeper to validate that every deployed Helm chart includes an accompanying, validated NetworkPolicy.
- **Egress Restriction:** Prevent compromised pods from initiating unauthorized outbound internet connections.

Limitations / Identified Research Gaps

Static manifest analysis cannot detect runtime dynamic network connections (e.g., dynamic external API endpoints). It must be paired with runtime eBPF telemetry to catch drift.

Paper 6 — Partially Trusting the Service Mesh Control Plane

Paper Details

- **Title:** Partially Trusting the Service Mesh Control Plane
- **Authors:** Constantin Adam, Abdulhamid Adebayo, Hubertus Franke, Edward Snible
- **Affiliation:** IBM T.J. Watson Research Center & Red Hat, USA
- **Publication / Venue:** IEEE Cloud / Workshop on Cloud Security, 2024
- **Domain:** Service Mesh Control Plane Security, Verifiable Configuration, Istio / Envoy Hardening

Problem Addressed

Existing Zero-Trust Service Mesh architectures place absolute, unquestioned trust in the service mesh control plane (e.g., istiod). In multi-tenant environments, managed Kubernetes offerings, or multi-cloud infrastructures, the application owner does not control the cluster infrastructure. A rogue cluster administrator or a compromised control plane can silently alter Envoy proxy routing rules, inject malicious telemetry sidecars, disable mTLS, or route sensitive application traffic to an attacker's inspection pod without the application owner's knowledge.

Approach / Methodology & Architecture

The authors introduce the concept of 'Partially Trusted Control Planes' using Verifiable Configurations. Application owners specify and cryptographically sign their desired service mesh configurations (e.g., routing tables, mTLS policies, authorization rules) using private asymmetric keys outside the cluster. The Envoy data plane proxy is enhanced with an internal cryptographic verification engine. When the control plane pushes configuration updates via xDS protocols, the proxy verifies the digital signature against the application owner's public key before applying any changes, effectively neutralizing rogue control plane commands.

What We Understood (Key Findings & Insights)

- **Neutralizing Insider Threats:** Privileged cluster admins and compromised control planes cannot inject unauthorized routing rules or disable mTLS.
- **Cryptographic Signature Verification in Envoy:** Using Ed25519 signatures, proxy verification overhead was measured at under 0.8 ms per configuration update.
- **Zero Impact on Data Plane Latency:** Signature verification occurs strictly during control plane configuration ingestion; runtime HTTP request forwarding latency is completely unaffected (0% overhead).
- **Dual-Control Governance:** Establishes a dual-control security model where infrastructure operators manage deployment resources, but application owners retain cryptographic governance over security policies.

Relevance to Our Project (TrustMesh)

This paper provides the exact architectural solution for TrustMesh's control plane integrity! In TrustMesh, security policies must remain untampered even if the underlying cloud infrastructure or managed Kubernetes control plane is compromised.

What We Plan to Apply

- **Signed Policy-as-Code:** Implement digital signing for all OPA Rego policies and Istio AuthorizationPolicies using Sigstore/Cosign.

- **Admission Verification:** Verify policy cryptographic signatures at Kubernetes admission before applying configurations to the cluster.

Limitations / Identified Research Gaps

The mechanism assumes the Envoy proxy process itself cannot be tampered with in memory on the worker node. If an attacker possesses root access on the worker node OS, they could potentially manipulate proxy memory—which leads directly to the hardware enclave solution in Paper 8.

Paper 7 — Performance Analysis of Zero-Trust Multi-Cloud

Paper Details

- **Title:** Performance Analysis of Zero-Trust Multi-Cloud
- **Authors:** Simone Rodigari, Donna O'Shea, Pat McCarthy, Martin McCarry, Sean McSweeney
- **Affiliation:** Munster Technological University & Pilz Ireland Industrial Automation Ltd., Cork, Ireland
- **Publication / Venue:** IEEE International Conference on Cloud Networking / Communications, 2024
- **Domain:** Multi-Cloud Performance Benchmarking, Istio Service Mesh, mTLS Latency & Resource Overhead

Problem Addressed

Enterprise organizations frequently cite performance degradation, latency jitter, and cloud resource bloat as the primary deterrents to adopting Zero-Trust Service Meshes across multi-cloud environments. Despite theoretical discussions, there was a complete absence of rigorous, empirical benchmark studies quantifying the actual performance trade-offs of enabling Zero-Trust mTLS across multiple distinct commercial cloud providers.

Approach / Methodology & Architecture

The authors built a standardized multi-cloud experimental testbed spanning Amazon Elastic Kubernetes Service (AWS EKS) and Google Kubernetes Engine (GKE). They deployed microservices across both clouds and executed rigorous comparative load tests under sequential and burst traffic patterns using Fortio. They measured Round Trip Time (RTT), latency percentiles (p50, p90, p99), latency standard deviation, sidecar memory consumption, and control plane CPU utilization across four scenarios: baseline HTTP without mesh, mesh with plaintext, mesh with PERMISSIVE mTLS, and mesh with STRICT mTLS.

What We Understood (Key Findings & Insights)

- **Reduced Latency Jitter with Istio:** Remarkably, Istio STRICT mTLS significantly reduced latency variability (standard deviation dropped from 34.6 ms down to 12.2 ms on sequential requests) due to Envoy's persistent HTTP/2 connection pooling and multiplexing.
- **Baseline Latency Cost:** The median (p50) latency increase introduced by mTLS encryption was only 1.8 ms to 3.2 ms across cloud regions, proving that encryption overhead is well within acceptable enterprise thresholds.
- **Memory Overhead is the Dominant Cost:** Envoy sidecar proxies consume approximately 50 MB to 90 MB of RAM per pod, which can become significant in microservice deployments with hundreds of replicas.
- **Cloud Provider Discrepancies:** Network performance and sidecar efficiency varied slightly between AWS EKS and GKE due to differences in underlying virtualized networking stacks (AWS VPC CNI vs. GKE Datapath V2).

Relevance to Our Project (TrustMesh)

Provides TrustMesh with the exact empirical benchmarking methodology, testing workflow, and performance baseline needed for Phase 7 (Testing & Empirical Evaluation). It also validates our multi-cloud deployment across AWS EKS and Azure AKS.

What We Plan to Apply

- **Benchmark Metrics:** Adopt Fortio/k6 testing workflows measuring p50, p95, p99 latency and standard deviation.
- **Connection Pooling Optimization:** Tune Envoy HTTP/2 connection pooling settings to minimize latency variance.
- **Resource Quota Planning:** Allocate appropriate sidecar memory limits (75-100 MB) during cluster capacity planning.

Limitations / Identified Research Gaps

The study only evaluated sequential and low-concurrency burst HTTP traffic; it did not evaluate massive gRPC streaming or sidecarless service mesh architectures (such as Istio Ambient Mesh).

Paper 8 — S-ZAC: Hardening Access Control of Service Mesh Using Intel SGX for Zero Trust in Cloud

Paper Details

- **Title:** S-ZAC: Hardening Access Control of Service Mesh Using Intel SGX for Zero Trust in Cloud
- **Authors:** Changhee Han, Taehun Kim, Woomin Lee, Youngjoo Shin
- **Affiliation:** School of Cybersecurity, Korea University, Seoul, Republic of Korea
- **Publication / Venue:** Electronics (MDPI), Vol. 13, Issue 16, Article 3213, August 2024. DOI: 10.3390/electronics13163213
- **Domain:** Trusted Execution Environments (TEE), Intel SGX, Access Control Hardening, Service Mesh Root of Trust

Problem Addressed

In public cloud environments, sidecar proxies execute access control enforcement and store TLS private keys in user-space memory on shared host nodes. Under the Zero-Trust principle, even the cloud infrastructure provider must be treated as untrusted. A privileged adversary (such as a malicious cloud insider, compromised hypervisor, or root-level host kernel exploit) can inspect or alter proxy process memory, steal private keys, or bypass access control checks entirely, rendering software-only Zero-Trust ineffective.

Approach / Methodology & Architecture

The authors propose S-ZAC, a hardware-hardened access control framework for service meshes utilizing Intel Software Guard Extensions (SGX). S-ZAC isolates the access control policy engine, policy store, and cryptographic key material inside an SGX hardware enclave (Trusted Execution Environment - TEE). S-ZAC utilizes SGX Remote Attestation to cryptographically verify enclave integrity to the control plane before receiving access policies, ensuring that unauthorized hosts cannot impersonate legitimate proxies.

What We Understood (Key Findings & Insights)

- **Hardware-Rooted Isolation:** Access control decisions and TLS private keys remain encrypted in hardware memory, shielded even from host root and hypervisors.
- **Minimal Enclave Transition Overhead:** Through careful memory management and asynchronous enclave calls (using Gramine-SGX), policy evaluation inside SGX added only 0.12 ms to request latency.
- **Scalability with Policy Volume:** S-ZAC maintains linear scalability, evaluating 100,000 policy rules in 14.8 ms within the protected enclave.
- **Remote Attestation for Zero Trust:** Guarantees that access control policies are distributed exclusively to proven, untampered proxy enclaves.

Relevance to Our Project (TrustMesh)

Demonstrates the pinnacle of Zero-Trust defense: hardware-rooted trust. While TrustMesh primarily targets software-level Kubernetes deployments, S-ZAC provides the architectural template for our High-Assurance TrustMesh extension for mission-critical banking and defense workloads.

What We Plan to Apply

- **Remote Attestation Concepts:** Incorporate cryptographic node and workload attestation into our SPIRE identity deployment.

- **Memory Protection Best Practices:** Ensure secrets and private keys in TrustMesh are locked in memory (mlock) and never written to disk.

Limitations / Identified Research Gaps

Requires Intel SGX-compatible cloud hardware (e.g., Azure DC-series or AWS Confidential VMs), increasing compute costs and limiting general portability across commodity Kubernetes nodes.

Paper 9 — Zero Trust Architecture in Cloud-Native Environments: Implementation Strategies & Best Practices

Paper Details

- **Title:** Zero Trust Architecture in Cloud-Native Environments: Implementation Strategies & Best Practices
- **Author:** Saurabh Verma (Amazon Web Services Inc., Columbus, Indiana, USA)
- **Publication / Venue:** International Journal of Computer Trends and Technology (IJCTT), Vol. 73, Issue 4, pp. 102-107, April 2025. DOI: 10.14445/22312803/IJCTT-V73I4P114
- **Domain:** Cloud-Native Security, AWS Architecture, Kubernetes Governance, Secrets Management

Problem Addressed

As enterprise workloads migrate to containerized, cloud-native platforms, the sheer dynamism of microservices creates severe security challenges: expanded ephemeral attack surfaces, identity sprawl across thousands of service accounts, static secrets embedded in code, and compliance gaps. Engineering teams lack unified, practical implementation blueprints for mapping theoretical Zero-Trust principles into production cloud-native toolchains.

Approach / Methodology & Architecture

The author synthesizes architectural strategies and production best practices derived from enterprise cloud-native implementations at AWS. The paper outlines a comprehensive layered defense: (1) Identity governance utilizing IAM Roles for Service Accounts (IRSA) and OpenID Connect; (2) Kubernetes admission control using Kyverno and OPA Gatekeeper; (3) Automated dynamic secrets management using AWS Secrets Manager with KMS envelope encryption; (4) Network isolation via Amazon VPC Lattice and service meshes; and (5) CI/CD pipeline scanning and Post-Quantum Cryptography (PQC) readiness.

What We Understood (Key Findings & Insights)

- **Least-Privilege Workload Scoping:** IRSA eliminates node-level IAM credentials by binding AWS IAM roles directly to specific Kubernetes Service Accounts via OIDC federated tokens.
- **Shift-Left Policy-as-Code:** Enforcing security policies at the admission controller stage prevents misconfigured manifests from ever reaching the etcd cluster state.
- **Automated Secret Rotation:** Hardcoded secrets must be replaced with dynamic secrets rotated automatically every 30 days or less using envelope encryption.
- **Post-Quantum Readiness:** Transitioning cryptographic algorithms to hybrid post-quantum schemes (e.g., ML-KEM) to defend against 'harvest now, decrypt later' attacks.

Relevance to Our Project (TrustMesh)

Provides direct, industry-validated implementation guidelines for TrustMesh's AWS EKS deployment foundation. The specific guidance on IRSA configuration and OPA admission controllers directly shapes Phase 3 and Phase 4 of our roadmap.

What We Plan to Apply

- **AWS IRSA Implementation:** Configure TrustMesh on AWS EKS using IRSA and OIDC federation for all control plane daemons.
- **Admission Controller Validation:** Deploy OPA Gatekeeper constraints validating pod security standards (non-root execution, read-only root filesystems).

- **Automated Secrets Management:** Integrate Kubernetes External Secrets Operator with cloud secrets stores.

Limitations / Identified Research Gaps

The paper focuses heavily on AWS-proprietary services (VPC Lattice, Secrets Manager, KMS), offering less detail on multi-cloud interoperability with Azure AKS or open-source neutral tooling like SPIRE.

Paper 10 — Zero-Trust Based Dynamic Access Control for Cloud Computing

Paper Details

- **Title:** Zero-Trust Based Dynamic Access Control for Cloud Computing
- **Authors:** Ri Wang, Chen Li, Kun Zhang, Bibo Tu
- **Affiliation:** State Key Laboratory of Mathematical Engineering and Advanced Computing, Zhengzhou, China
- **Publication / Venue:** Cybersecurity (Springer Open), Vol. 8, Article 12, February 2025. DOI: 10.1186/s42400-024-00320-x
- **Domain:** Dynamic Risk-Based Access Control, Reinforcement Learning, Continuous Trust Evaluation

Problem Addressed

Traditional access control models—such as Role-Based Access Control (RBAC) and Attribute-Based Access Control (ABAC)—rely on static rules matched against dynamic metrics. Once a user or workload is authenticated, permissions remain static throughout the session. If an entity's credentials are stolen, or if a workload begins exhibiting malicious behavior (such as mass data exfiltration or abnormal API requests), static access control cannot respond in real time, causing catastrophic damage during the delayed response window.

Approach / Methodology & Architecture

The authors propose DR-TBAC (Dynamic Risk and Trust-Based Access Control), a real-time Zero-Trust access control framework for cloud computing environments. DR-TBAC combines continuous risk assessment with dynamic trust evaluation. It integrates a Multi-Agent Deep Deterministic Policy Gradient (MADDPG) and Bidirectional Gated Recurrent Unit (BiGRU) neural network model to analyze multidimensional behavioral attributes (request frequency, IP reputation, resource sensitivity, time-of-day, and access history) in real time. Permissions are dynamically adjusted during active sessions based on real-time risk scores.

What We Understood (Key Findings & Insights)

- **Superior Anomaly Detection Accuracy:** The MADDPG-BiGRU model achieved 0.98 precision, 0.97 recall, and 0.98 F1-score, significantly outperforming traditional SVM (0.83), Autoencoders (0.88), and static RBAC (0.60 F1-score).
- **Continuous In-Session Authorization:** Access decisions are not one-time gates; access permissions are continuously recalculated during ongoing transactions.
- **Real-Time Evaluation Latency:** Online risk evaluation completed in 4.8 ms per request, proving feasibility for high-speed cloud transaction processing.
- **Adaptive Privilege Adjustment:** When anomaly scores rise, the system does not simply terminate sessions; it dynamically downgrades permissions (e.g., from write to read-only) or triggers step-up verification.

Relevance to Our Project (TrustMesh)

Provides the mathematical and algorithmic foundation for TrustMesh's Continuous Verification Engine! It enables TrustMesh to evolve beyond static OPA allow/deny rules into a risk-adaptive control plane that continuously monitors workload transaction patterns.

What We Plan to Apply

- **Dynamic Trust Scoring:** Incorporate request frequency and deviation metrics into TrustMesh trust score algorithms.
- **Adaptive Authorization Enforcement:** Configure TrustMesh PEPs to dynamically downgrade privileges when anomalous request rates occur.

Limitations / Identified Research Gaps

Training deep reinforcement learning models requires substantial computational resources and baseline telemetry datasets, which may be challenging to bootstrap in brand-new clusters without historical data.

12. Synthesis: 10 Technical Decisions Taken Forward into TrustMesh

Grounding our engineering choices directly in the empirical findings of these 10 papers, we have established ten technical architectural decisions that govern the development of TrustMesh:

- **1. 100% Secretless Inception:** Eliminate all static API keys, service account tokens, and database passwords from container manifests; bootstrap exclusively via cloud-native managed identities and OIDC (Paper 1).
- **2. Multi-Cloud Verifiable Identity Federation:** Implement cross-cloud workload attestation using decoupled verifiable claims rather than proprietary single-cloud IAM (Paper 2).
- **3. Vulnerability-Driven Mesh Access (VDT):** Continuously correlate container SBOMs and CVE scores with Istio AuthorizationPolicies to dynamically quarantine vulnerable pods (Paper 3).
- **4. Strict PDP / PEP Decoupling:** Centralize policy decisions in an asynchronous OPA control plane while distributing enforcement at sidecar proxies and admission gates (Paper 4).
- **5. Automated NetworkPolicy Generation:** Deploy default-deny Kubernetes NetworkPolicies across all namespaces and automatically infer least-privilege egress/ingress rules (Paper 5).
- **6. Cryptographically Signed Verifiable Policy-as-Code:** Sign all OPA Rego policies and Istio routing manifests using Sigstore/Cosign, verifying signatures before admission (Paper 6).
- **7. Connection Pooling for Latency Optimization:** Leverage Envoy HTTP/2 connection pooling to exploit Istio's proven capability to reduce latency jitter under sustained load (Paper 7).
- **8. Hardware Enclave (SGX) High-Assurance Architecture:** Adopt enclave-hardened access control principles for cryptographic key protection in mission-critical deployments (Paper 8).
- **9. AWS EKS IRSA and Admission Governance:** Standardize on AWS IAM Roles for Service Accounts and OPA Gatekeeper admission validation for non-root execution (Paper 9).
- **10. Continuous Risk-Adaptive Session Authorization:** Evolve authorization from static boolean checks into continuous risk-weighted evaluations based on request behavior and drift (Paper 10).

13. Comprehensive Bibliography & References

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